

Bandlapalli Roshan Babu

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Professional Summary

M.Tech in Artificial Intelligence & Data Science with a strong foundation in Deep Learning, NLP, and Quantum Machine Learning. Proven ability to design and implement end-to-end AI solutions, from concept to deployment. Currently pioneering the development of a novel Quantum-Classical Facial Biometric System. Selected as a Qiskit Advocate (2025) and active in promoting AI literacy as a Perplexity Campus Ambassador. Seeking roles as an AI Engineer, Research Scientist, or Quantum ML Engineer in the tech industry.

Technical Skills

- **Programming Languages:** Python, Java, SQL, JavaScript
- **AI/Deep Learning Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Core AI Competencies:**
 - **Deep Learning:** CNNs, RNNs, Autoencoders, Transformers, GANs, YOLO
 - **Natural Language Processing (NLP):** Tokenization, Sentiment Analysis, BiLSTM, GloVe Embeddings, Transformer Models, LangChain, LangGraph
 - **Machine Learning:** Classification, Regression, Clustering, Anomaly Detection
- **Quantum Computing:** Qiskit, Quantum Circuits, Hybrid Quantum-Classical Models, QCNN, QSVM, Variational Quantum Circuits, QASM Simulator
- **Data Processing & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy
- **Tools & Platforms:** Git, Jupyter Notebook, Google Colab, VS Code, Streamlit, Anaconda, Ubuntu, IBM Quantum Experience

Education

Mahindra University, Hyderabad **Aug 2024 – Present**
M.Tech, Artificial Intelligence & Data Science
CGPA: 7.38/10.0 — Awarded Merit-Based Scholarship

Presidency University, Bangalore **Jun 2020 – May 2024**
B.Tech, Computer Science & Engineering (AI & ML)
CGPA: 8.51/10.0

Professional Experience & Research

Research Intern — Lloyd's Bank @ Mahindra University, Hyderabad **Aug 2025 – Present**

- Spearheading the development of the first-ever **Quantum Facial Biometric System**, a groundbreaking project integrating classical Computer Vision with Quantum Machine Learning (QML).
- Designing and implementing hybrid quantum-classical pipelines to enhance the security and efficiency of facial recognition.
- *Technologies: Qiskit, Python, PyTorch/TensorFlow, Computer Vision libraries*

Perplexity Campus Ambassador (Volunteer) **Aug 2025 – Present**

- Promote AI literacy and drive the adoption of Perplexity AI tools among the student community.
- Organize and lead sessions to educate peers on leveraging AI for research and development.

Machine Learning Intern — Bharat Intern **Oct 2023 – Nov 2023**

- Developed end-to-end machine learning models for House Price Prediction, Wine Quality Prediction, and Iris Flower Classification.

- Engineered data preprocessing pipelines and validated model performance, sharing results and code via GitHub.

Machine Learning Internship Training — Skill Vertex

May 2023 – Jun 2023

- Completed intensive training in supervised and unsupervised learning algorithms.
- Gained hands-on experience in regression, classification, and clustering using Python and Scikit-learn on real-world datasets.

Projects

Quantum-Classical Pipeline for Facial Biometrics (Research Project)

- **Objective:** To develop a novel facial recognition system that leverages the computational advantages of quantum circuits for enhanced feature extraction or classification.
- **Contributions:** Designing variational quantum circuits (VQCs) to work in tandem with classical convolutional neural networks (CNNs). Experimenting with Quantum Convolutional Neural Networks (QCNNs) and Quantum Support Vector Machines (QSVM) on biometric data.
- **Technologies:** Qiskit, Python, OpenCV, PyTorch

EDA on IPL Dataset

- Performed exploratory data analysis (EDA) on IPL cricket dataset to uncover trends in team performance, batting, and bowling statistics.
- Created data-driven visualizations to support insights for player and match analysis.

Plant Prediction using ML (Crop Recommendation System)

- Built a machine learning model to predict the best crop for a given state, season, and month with high accuracy.
- Improved agricultural decision-making by leveraging climate and soil parameters.

Human-in-the-Loop Generative AI Application (LLMs)

- Built an interactive AI application allowing users to guide and correct AI-generated outputs. Implemented a robust backend for managing dynamic AI workflows.
- **Technologies:** LangChain, Streamlit, Python

QA Agent with LangGraph

- Designed and built an intelligent query-answering agent utilizing Large Language Models (LLMs) and the LangGraph framework for complex reasoning and information retrieval tasks.

Multilingual Customer Support Chatbot (NLP)

- Engineered a transformer-based chatbot capable of understanding and responding in multiple languages.
- **Technologies:** Transformers, Python, Streamlit

Emotion Detection using GloVe-BiLSTM

- Implemented a sophisticated NLP pipeline using GloVe embeddings and a Bidirectional LSTM (BiLSTM) network for multi-class emotion recognition from text, achieving high classification accuracy.

Anomaly Detection with Autoencoders on ECG Data

- Developed an unsupervised anomaly detection system using Autoencoders on the ECG5000 dataset. The model, trained exclusively on normal samples, used reconstruction error (MAE) to identify

anomalies with high accuracy.

Denoising Autoencoder for MNIST Image Reconstruction

- Designed a convolutional autoencoder in Keras to remove Gaussian noise from MNIST images.
- Successfully reconstructed clean images from noisy inputs, improving image quality in test samples.

Combined Cycle Power Plant Energy Output Prediction using ANN from Scratch

- Implemented a custom Artificial Neural Network from scratch using only NumPy, including forward/backward propagation, mini-batch gradient descent, and L2 regularization.
- Achieved an 18% reduction in forecast error.

Certifications

- AICTE Certified FDP on Quantum Computation (2025)
- Qiskit Advocate Program (2025)
- IBM: Quantum Machine Learning — Basics of Quantum Information — Fundamentals of Quantum Algorithms
- Kaggle: Intro to Machine Learning (2023)
- Meta-Coursera: Principles of UX/UI Design — Programming with JavaScript — Introduction to Front-End Development
- Internshala: Programming with Python (2022)
- IBM: Introduction to Artificial Intelligence (2021)

Awards & Achievements

- **Publication:** Published a paper on "Inventory Management using AI" in ICIET
- **Qiskit Advocate:** Selected for the prestigious Qiskit Advocate Program (2025)
- **Google Code Vipasana 2025:** Participated in the prestigious Google coding competition
- **Sports Leadership:** Incharge and Captain of the University Throwball Club, leading from the front
- **Sports Achievement:** Represented the Indian Men's Throwball Team as a Player in the Indo-Sri Lanka Youth Cup 2024 (Runner-up)
- **Creative Arts:** Participated as the Director of Photography (DOP) in a regional short film competition

Hobbies & Interests

- Playing chess, volleyball, badminton, and throwball
- Traveling and exploring new cultures
- Watching movies and sharing reviews/opinions
- Reading newsletters and staying updated on the latest tech